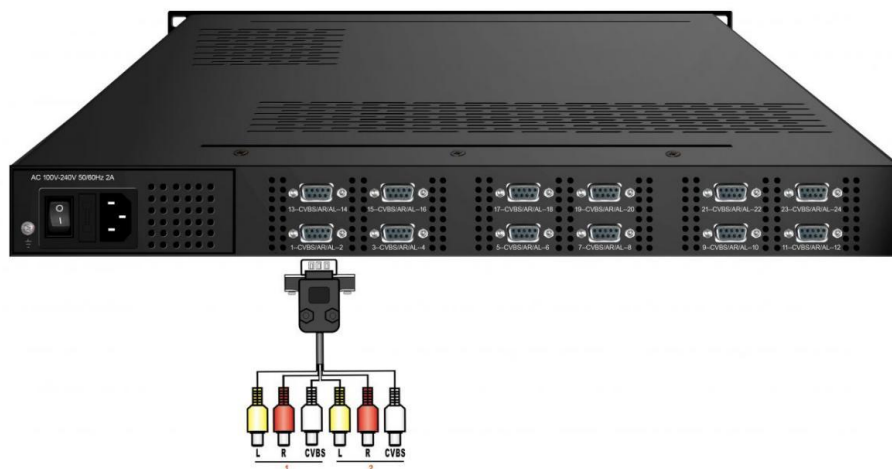


FMUSER FUQ-4653H-HA Encoder Modulator 8-24 CVBS Inputs 4 RF Outputs DVB-C-T-ISDBT MPEG-2 Scrambling Web Management for Hotel IPTV and CATV 8/12/24* CVBS/1 DVB-C/T/(T)/ISDBT Tuner
 —> 4* DVB-C/T Digital RF



DVB-T RF out front Panel



OSD (Logo/PPT/Caption/QR Code) Insertion



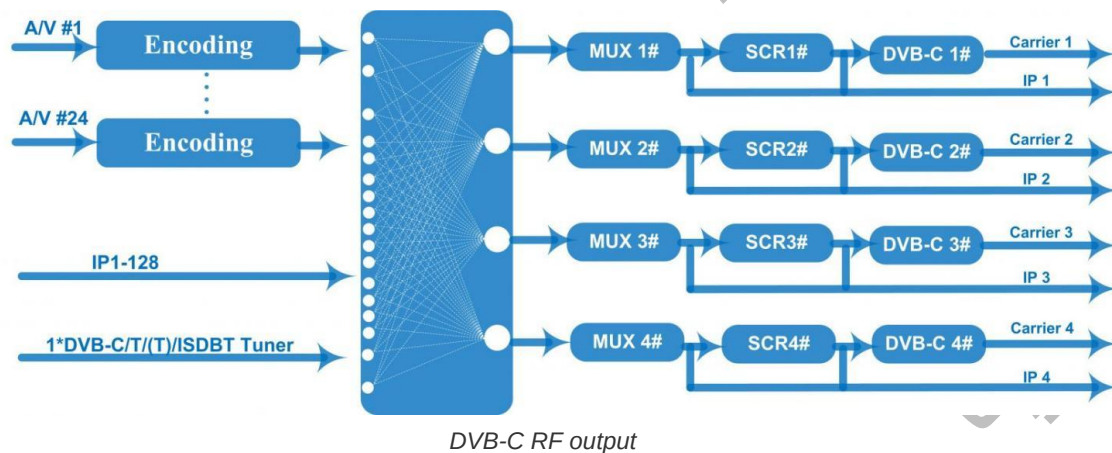
Product Overview

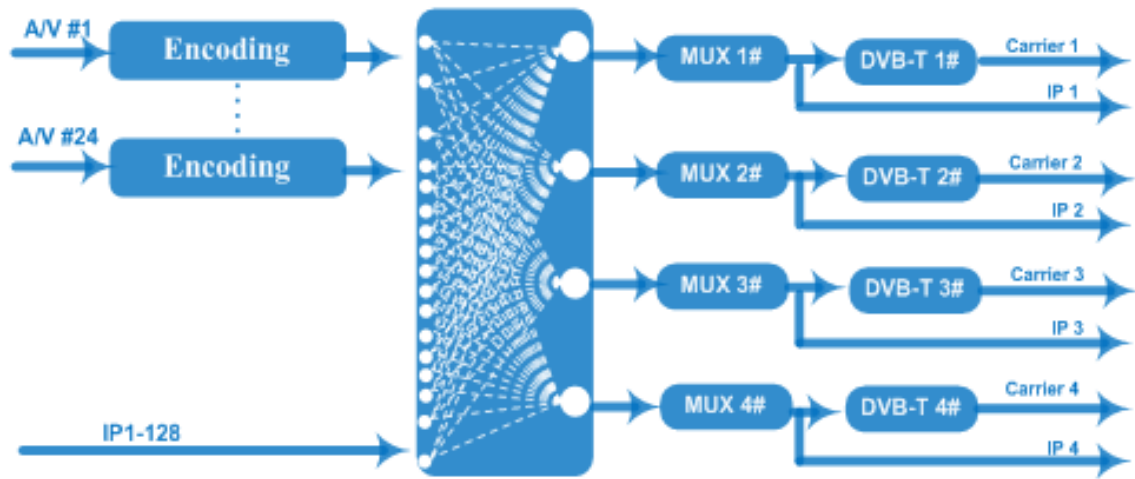
The FUQ-4653H/HA combines MPEG-2 encoding, multiplexing, and DVB-C/T/ISDBT modulation in a 1RU chassis. It processes 8-24 CVBS inputs, 128 IP streams, and delivers 4 RF outputs with adjustable levels. This web-managed solution supports PSI/SI editing and ETSI-compliant scrambling. Ideal for hotel TV systems, hospital networks, and small-scale CATV operators.

Key feature

Broadcast Security: ETSI-compliant scrambling (H/HA models) and OSD/QR code insertion (HA model) for content control.

Principle Chart





DVB-T RF output

Technical specification

Input	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45	8/12/16/20/24 CVBS inputs , RCA interface , 1 DVB-C/T/(T)/ISDBT Tuner for remux, F type interface (only for DVB-C RF out) 128 IP input over UDP and RTP, DATA1and DATA2,RJ 45
Tuner Section	DVB-C	Standard	Standard	Standard	J.83A(DVB-C), J.83B, J.83C	J.83A(DVB-C), J.83B, J.83C
Tuner Section	DVB-C	Frequency In	Frequency In	Frequency In	60 MHz~890 MHz	60 MHz~890 MHz
Tuner Section	DVB-C	Constellation	Constellation	Constellation	16/32/64/128/256 QAM	16/32/64/128/256 QAM

Tuner Section	DVB-T/(T)	Frequency In	Frequency In	Frequency In	60 MHz~890 MHz	60 MHz~890 MHz
Tuner Section	DVB-T/(T)	Bandwidth	Bandwidth	Bandwidth	6/7/8 M bandwidth	6/7/8 M bandwidth
Tuner Section	ISDB-T	Frequency In	Frequency In	Frequency In	60 MHz~890 MHz	60 MHz~890 MHz
Video	Resolution	720×480_60i, 544×480_60i, 352×480_60i 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i
Video	Resolution	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i 352×200_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i 352×200_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i 352×200_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i 352×200_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i 352×200_50i, 176×288_50i, 176×144_50i
Video	Encoding	MPEG-2	MPEG-2	MPEG-2	MPEG-2	MPEG-2
Video	Bit-rate	0.5Mbps~8 Mbps each channel	0.5Mbps~8 Mbps each channel	0.5Mbps~8 Mbps each channel	0.5Mbps~8 Mbps each channel	0.5Mbps~8 Mbps each channel
Video	Rate Control	CBR, VBR	CBR, VBR	CBR, VBR	CBR, VBR	CBR, VBR
Video	GOP Structure	GOP_0_B, GOP_1_B, GOP_2_B, GOP_3_B	GOP_0_B, GOP_1_B, GOP_2_B, GOP_3_B	GOP_0_B, GOP_1_B, GOP_2_B, GOP_3_B	GOP_0_B, GOP_1_B, GOP_2_B, GOP_3_B	GOP_0_B, GOP_1_B, GOP_2_B, GOP_3_B
Video	Advanced Pretreatment	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction
Audio	Encoding	MPEG-1 Layer 2, AC3 (2.0)	MPEG-1 Layer 2, AC3 (2.0)	MPEG-1 Layer 2, AC3 (2.0)	MPEG-1 Layer 2, AC3 (2.0)	MPEG-1 Layer 2, AC3 (2.0)

Audio	Sampling rate	48KHz	48KHz	48KHz	48KHz	48KHz
Audio	Resolution	24-bit	24-bit	24-bit	24-bit	24-bit
Audio	Bit-rate	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel
Multiplexing	Maximum PID Remapping	180 input per channel	180 input per channel	180 input per channel	180 input per channel	180 input per channel
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Scrambling (Only for DVB-C RF out)	Maximum simulcrypt CA	4	4	4	4	4
Scrambling (Only for DVB-C RF out)	Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling (Only for DVB-C RF out)	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	DVB-C	RF out	RF out	4*RF DVB-C out (4 carriers combined output)	4*RF DVB-C out (4 carriers combined output)	4*RF DVB-C out (4 carriers combined output)

Modulation	DVB-C	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	MER	MER	≥40db	≥40db	≥40db
Modulation	DVB-C	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	-25 ~ -1 dBm, 1dBm step	-25 ~ -1 dBm, 1dBm step	-25 ~ -1 dBm, 1dBm step
Modulation	DVB-C	Symbol Rate	Symbol Rate	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping	5.0Msps~7.0Msps, 1ksps stepping
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	16/32/64/128/256QAM
Modulation	DVB-C	Constellation	Constellation	J.83A	J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	8M	8M	6M
Modulation	DVB-T	RF out	RF out	4*RF DVB-T out (4 carriers combined output)	4*RF DVB-T out (4 carriers combined output)	4*RF DVB-T out (4 carriers combined output)
Modulation	DVB-T	Standard	Standard	EN300744	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	2K,	2K,	2K,
Modulation	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32

Modulation	DVB-T	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF output level	RF output level	-28~ -3 dBm, 1 dBm step	-28~ -3 dBm, 1 dBm step	-28~ -3 dBm, 1 dBm step
Stream output	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)	RF output (F type interface)
Stream output	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface	4 IP MPTS output over UDP/RTP, 1*1000M Base-T Ethernet interface
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System function	English language	English language	English language	English language	English language	English language
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×410mm×44mm	482mm×410mm×44mm	482mm×410mm×44mm	482mm×410mm×44mm
Miscellaneous	Environment	Environment	0~45℃(work)■-20~80℃■Storage■	0~45℃(work)■-20~80℃■Storage■	0~45℃(work)■-20~80℃■Storage■	0~45℃(work)■-20~80℃■Storage■
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz

Order Information

	FU-4653H (DVB-C RF out)	FU-4653H (DVB-T RF out)	FU-4653HA (DVB-C RF out)	FU-4653HA (DVB-T RF out)
Scrambling Function	√		√	
DVB-C/T/(T)/ISD BT Tuner Input	√		√	
OSD Insertion Function			√	√

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)